Introduction

Here we introduce the importance of brain ventricles segmentation.
Here we talk about the noise impact into MRI images (dovrebbe essere importante, se trovi qualcosa mettici una frase)
Here we declare that **our goal** was to develop a workflow for brain ventricles segmentation, easy to change - to adapt it to different images - and quick in showing results of user modifications.

Materials and methods

We implemented our workflow as a MatLab mlx in order to reach our goals.
We used two three-dimensional MRI images to test our workflow: a T1-weighted one and a T2-weighted one.
We split the mlx into sections: some of them for image pre-processing, some of them for the actual segmentation and some of them for the results evaluation.

Table of Contents

Inspired by …(document_ref)… we tried to distinctly segment the two ventricles, calculating the respective volumes
Then, we compared the performances of our workflow on same images with and without noise.

Pre-processing
Our workflow was made to offer three pre-processing methods: Noise-removal, Normalization and Gamma-adjusting.

Area computation
After the selection of a region of the image through a double-click, the workflow show the selected region and calculates its area.

Volume computation
Every-time a region is selected, it can be 'added' to a three-dimensional logical volume, where all the regions chosen from the various slices of the MRI volume are stored into respective logical slices. Finally, it's possible to know the volume measure of the selected volume.

Performance evaluation
The workflow calculates Sensitivity, Specificity and Dice correlation of the computed segmentation, with respect to a manual segmentation uploaded by the user.

Results

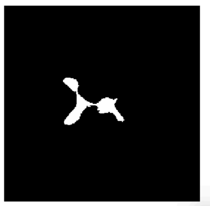
In order to measure the quality of the workflow and of our usage of it, we manually segmented the ventricles in slice 16 of the T1

Manual segmentation **Computed segmentation**

```
sensitivity_on_restricted_region = 0.7568
specificity_on_restricted_region = 0.9748
dice_coeff_on_restricted_region = 0.8413
```

image, then we segmented them in the same slice through the workflow and finally calculated Sensitivity, Specificity and Dice-correlation:
these are the values obtained over a restricted portion of the image (otherwise the Specificity would have been obviously 0.99).

We managed to complete the volume calculation over the T1 MRI, but when we tried over t2 and understood our workflow is still too simple:



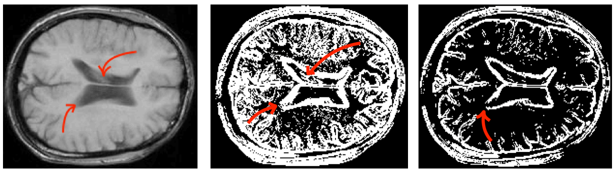
this is the type of region we found over T2 slice

We confronted results over slice 16 of T1 with results obtained applying noise simulation.
With noise we obtained this:


Gaussian, mean 0,005 mean 0,01

With salt&pepper we managed to have nice results even with density 0,02:

Discussion
Our best achievement were obtained on T1 images, as expected since we mainly used them to build the workflow
We expected also our workflow to be more … to the salt&pepper noise rather than to gaussian noise: our goal was to segment and it becomes more difficult if each pixel is the average of its neighbours
Our main obstacles were zones of slow intensity-change. Here are reported two regions as example, with the possible imperfections they brought to segmentation:



On the left: the original image
Central: segmentation with threshold=0.1
Here it's possible to notice how the upper problematic region creates chaos in the border-computation system; and how the lower problematic region is still 'closed' but not as perfectly as the rest of the border.
On the right: segmentation with threshold=0.1
Here it's possible to notice how the lower problematic region has now lead border to be 'open': the area computation of the lower ventricle can't be applied using this threshold.